

HW#8**REF: READ THEORY MANUAL SECTIONS §2.3-§2.6
MODULE 8 – LECTURE SLIDES**

1. Reactor Theory Manual - Chapter 2 (page 2-31 §2.14), DO QUESTION #1
Hint: For #1 – reference §2.5 ‘Exponential Power Equations’. Use the InHour curve or ‘One-Group’ approximation to relate ρ to τ . Please show all work!

2. Define Reactor Period ‘ τ ’.

3. Given the “One-Group Approximation” to the InHour Equation (Eq. #2.10), if $\lambda_{\text{eff}} \sim 0.1 \text{ sec}^{-1}$ (approximation for “slightly critical”) & $\beta_{\text{eff}} = 730 \text{ pcm}$:
 - a) What is the period τ (sec) if $\rho_{\text{net}} = 0.0030 \Delta k/k$ (300 pcm)?

 - b) What is the Startup Rate (SUR) in Decades-per-minute (DPM)? How long will it take for reactor power to increase by a factor of ten (10)?

4. From the InHour curve:
 - a) What is the period τ (sec) if $\rho_{\text{net}} = 0.0030 \Delta k/k$?

 - b) Using this period, how long would it take to increase reactor power from 1 W to 1 kW?

5. What is the difference (in seconds) for how long (t) it would take to increase power from 1 Watt to 1 kW using the two different periods from Problems #3a & #4a above?